MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $(2;1)\cdot(3;-4)$ equals

   A. 2
   B. 10
   C. $(6;-4)$
   D. -2

2. The norm of (-3; 4) is

   A. 7
   B. 25
   C. 5
   D. 1

3. Two vectors of zero product are

   A. equal
   B. collinear
   C. Opposite
   D. orthogonal

4. Norms 2 and 3, angle 60 $^\circ$: product

   A. 6
   B. 3
   C. $\sqrt{3}$
   D. 0

5. $\|u-v\|^2$ value

   A. $\|u\|^2-\|v\|^2$
   B. $\|u\|^2+\|v\|^2$
   C. $\|u\|^2-2u\cdot v+\|v\|^2$
   D. $\|u\|^2+2u\cdot v+\|v\|^2$

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. An obtuse angle between nonzero vectors gives a

   A. positive
   B. zero
   C. equal to 1
   D. Negative

2. In an orthonormal basis, $u\cdot i$ is

   A. the x-coordinate of u
   B. the norm of u

C. the y-coordinate of u

D. always zero

3. The locus $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ for $A \neq B$ is

A. the perpendicular bisector

B. the line AB

C. the circle with diameter AB

D. the midpoint alone

4. If $u \cdot v = 3$, $(2u) \cdot (-v)$ is

A. -3

B. -6

C. 6

D. 9

5. For $AB = AC = 2$ and a right angle at A, BC value

A. 4

B. 2

C. 8

D. $2\sqrt{2}$

True or false? Justify.

1. The scalar product of two vectors is always positive.

2. The zero vector is orthogonal to every vector.

3. $u \cdot v = 0$ implies that one of the vectors is zero.

Personal review

I distinguish norm and squared norm; I choose a suitable formula; I check the coordinate system; I write an equivalence for a locus.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.